



MANIPAL UNIVERSITY
JAIPUR



logitech

sarvam



MUJ HACKX 4.0

ढाल **Dhaal** — पैसे भेजने से पहले, **एक जाँच।**

TEAM ALL STARS

TEAM ID: [Registration ID]

PS: Consumer Protection Against Payment Scams · THEME: FINTECH (PS #7)

IDEA · SOLUTION

एक जाँच।

**One check
before you pay.**

paste it · photo the QR ·
or just SPEAK the call

Rules compute the verdict. AI only explains it — and can never change it. Never “safe”, only “no known risk”.

01 जाँच

Verdict + **every weighted signal**, spoken aloud. It asks your **intent** first — “expecting money IN” meeting a PAY-QR is caught with **zero scam words**.

02 परिवार की ढाल

Risky payment → family's 10-second **Allow/Block**. Risk shared, never transactions.

03 धोखों का नक्शा

Report → human verify → **everyone's live blacklist**. Burned in Jaipur ≠ works in Jodhpur.

04 पहला घंटा

1930 script + complaint + bank letter, auto-drafted in Hindi — the hour that gets money back.

LIVE: dhaal-delta.vercel.app

judges: check your own inbox in Q&A

TECHNICAL APPROACH

Verdict = deterministic code. AI narrates.

Next.js PWA · FastAPI (Vercel Python) · MongoDB Atlas · Claude sonnet-5 · Sarvam Saarikā ASR + Bulbul TTS · jsQR on-device



67 AUTOMATED CHECKS GREEN **5.4s** P50 CHECK, LIVE PROD **ms** VERDICT ITSELF (RULES)

USER EXPERIENCE & DESIGN

A public-safety poster, not a SaaS dashboard.

सुरक्षा-poster पहचान

Ink + saffron hazard language, Anek Devanagari display, a rubber-stamp community count on the verdict card.

अगले अरब users के लिए

हिं/EN toggle holds across every page and output —
incl. the spoken warning · voice in AND out · QR
decoded on-device (photo never leaves the phone) ·
entry-level phones.

इज़ात के साथ हिफ़ाज़त

Guardian sees the risk, never the transactions.
Risky checks ask a decision; clean ones are quiet notes.

Danger screams. Safety whispers.

HONEST LANGUAGE EVERYWHERE — “NO KNOWN RISK”, NEVER “SAFE”

FEASIBILITY & VIABILITY

Not a concept — **deployed tonight.**

51/60HELD-OUT BATTERY vs PROD
PUBLISHED UNEDITED**5.4s**P50 FULL CHECK
LIVE PROD**450+**REPORTS IN ATLAS
(SEEDED PILOT + LIVE)**₹0**INFRA — RUNS ON
FREE TIERS

Risks, already engineered away

API outage → deterministic fallbacks — the demo survives dead venue Wi-Fi.

False positives → context-gated rules; real bank OTP/debit SMS score **0**; misses published unedited in **docs/EVAL.md** — we grade ourselves before judges do.

Blocklist poisoning → human verification before any indicator goes live.

Scale → stateless serverless + Atlas; TTS caching; marginal cost per check ≈ a fraction of a paisa.

dhaal-delta.vercel.app**dhaal-api.vercel.app**

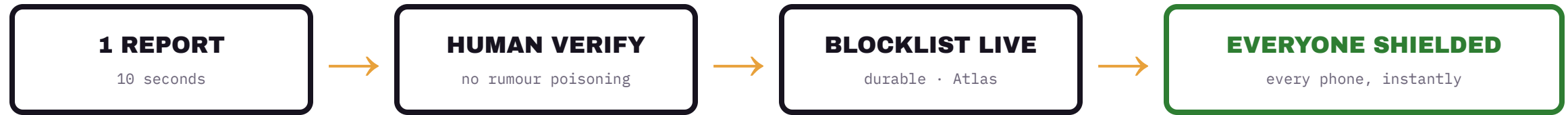
GUARDIAN + FLYWHEEL + VOICE LOOP VERIFIED END-TO-END

IMPACT & BENEFITS

₹22,495 Cr

lost to cyber fraud in 2025 —
and the victim pressed PAY themselves.

24.51B UPI TXNS/MONTH (NPCI AUG '26) · 28.1 LAKH COMPLAINTS (I4C 2025) · ₹805 CR UPI FRAUD FY26 (PARLIAMENT)



🛡️ **HERD IMMUNITY FOR FRAUD — THE SHIELD COMPOUNDS WITH EVERY USER**

Every Indian with a phone — free forever · the elderly — the most-targeted group, protected by family (guardian mode).

Banks & UPI apps — fewer reimbursements · cyber cells — structured early reports; fast 1930 reporting has already saved ₹7,130+ crore.

BUSINESS MODEL & SUSTAINABILITY

Citizens: **FREE forever.**

Protection this fundamental must not be paywalled — and the marginal cost per check is a **fraction of a paisa** by construction.

Institutions pay.

Risk-API + verified-blocklist licensing — banks, fintechs, UPI apps (they reimburse this fraud today).

Pre-payment SDK inside UPI apps — the check runs before the PAY button renders.

Deployment partnerships — state cyber cells & the CSC network (assisted mode).

The moat

The community blocklist **compounds with every user** — a network effect no new entrant starts with. Roadmap: one-tap **1930/I4C** hand-off · **10+ Indic languages** via the same Sarvam pipeline (one parameter) · auto-verify thresholds for moderation at scale.

RESEARCH & REFERENCES

Data sources (verified 11 Sep 2026)

NPCI monthly UPI statistics — 24.51B txns / ₹29.82 lakh crore, Aug 2026

I4C (MHA) — ₹22,495 crore lost · 28.1 lakh complaints (2025) · 32.4M calls to 1930

Ministry of Finance, Rajya Sabha reply — ₹805 crore UPI fraud, 10.64 lakh incidents, FY26

PIB / MHA — ₹7,130+ crore saved via CFCFRMS (1930) early reporting

RBI — guidelines on unauthorised-transaction liability

Domain research

Scam-script corpus from I4C advisories: digital arrest, KYC-expiry, collect-refund bait, OLX/army, job-task, loan-fee, coercion

upi:// deep-link collect-vs-pay semantics · NPCI PSP handle registry for VPA verification



dhaal-delta.vercel.app

github.com/SaudSatopay/MUJ-HACKX

हर report, सबकी ढाल

